Field Shift: Team Technical Guide

NASA Space Apps Challenge 2026 | Detailed plan for the whole team

1. Project in Brief

Problem: farmers face changing temperature, unpredictable rain, water shortage, and declining soil health, which makes it hard to choose good crop rotations. **Solution:** a decision-support tool that uses NASA Earth observations, local soil information, crop facts, and the farmer's own priorities to compare at least three rotation strategies (including the existing one) and explain each one, including its evidence, benefits, information gaps, and risks.

Raw data -> harmonize -> signals -> score rotations -> uncertainty + explanation -> app

Everything in this guide is a **proposed plan**. The team should confirm or change the choices (pilot area, datasets, crops, tools) in its first meeting.

2. Step 1: Pilot Area and Scope

We build and demonstrate the tool for **one small region** instead of the whole world. A suggested choice is a farming area in the Nile Delta, because it is close to us, has strong irrigation and water-stress issues, and we can reason about its crops. A pilot area is roughly 20 x 20 km or smaller.

What to decide

- The exact boundary of the area (a simple rectangle of coordinates is enough).
- Which 5 to 8 crops are common there (see the crop table in Step 6).
- The time period for historical data (suggested: 2015 to 2025, about 10 years, so trends are visible).

3. Step 2: Data Collection

We start with a small, reliable set of datasets. The resolutions below are approximate and should be confirmed on each dataset's page when downloading.

Dataset	Gives us	Approx. detail	Role in the tool
NASA POWER	Temperature, solar radiation, humidity, wind, precipitation (daily)	About 50 km squares	Heat and frost days, evaporation estimate, easiest to access (has a simple API)
GPM (IMERG)	Rainfall	About 10 km, every 30 min	Rainfall totals, variability, extreme rain, drought periods
SMAP	Soil moisture	About 9 to 36 km, every 2-3 days	Soil moisture anomalies, dry-spell detection
MODIS (and/or Landsat)	Vegetation index (NDVI)	MODIS about 250 m to 1 km; Landsat 30 m	Green and bare periods, crop stress, seasonal cycle
SRTM	Elevation	About 30 m	Slope for erosion risk, low-lying (frost-prone) spots
SoilGrids (not NASA)	Soil texture, organic carbon, pH	About 250 m	Local soil information (free global dataset)

Dataset	Gives us	Approx. detail	Role in the tool
GRACE / GRACE-FO (bonus)	Underground water storage change	Very coarse (hundreds of km), monthly	Long-term regional water trend only
ECOSTRESS (bonus)	Land surface temperature, plant water stress	About 70 m, irregular times	Heat stress hotspots if time allows

Practical notes

- Most NASA data needs a free **NASA Earthdata account**; create one early.
- Prefer tools that let us download only our small area (subsets) instead of huge global files.
- **Download and save the data before the demo.** Live downloads during the presentation are risky and slow.
- Keep a simple log of every dataset used: name, version, date range, and where it came from. Judges will ask.

4. Step 3: Harmonization (Making Datasets Comparable)

The datasets have different map sizes and update speeds, so they cannot be compared directly. We convert all of them to **one common grid** and **one monthly time step**.

Task	How
Choose the common grid	Pick one square size (suggested: about 1 to 10 km, depending on the coarsest dataset we rely on). Every dataset is converted to it.
Fine data to coarse grid	Detailed data (such as NDVI at 250 m) is averaged into the bigger squares.
Coarse data to fine grid	Coarse data (such as SMAP) is stretched across the smaller squares. This adds no new detail, so we record it as a limitation.
Choose the time step	Monthly. Frequent data is combined into one value per month.
Combine in the right way	Rainfall is **summed** into monthly totals. Temperature, soil moisture, and NDVI are **averaged**. Also keep monthly minimum and maximum temperature for heat and frost.
Handle gaps	Clouds can hide NDVI. Fill short gaps carefully (for example, from nearby months) and **flag every filled value** so uncertainty can reflect it.
Final result	One clean table: for every grid square and every month, values for rainfall, temperature, soil moisture, NDVI, plus slope and soil properties.

5. Step 4: Turning Data into Signals (Feature Engineering)

Raw numbers mean little to a farmer. We calculate **signals** that describe conditions and risks.

Signal	Meaning	Calculated from
Soil moisture anomaly	How much drier or wetter than normal for that month	Monthly soil moisture compared with its long-term average for the same month
Rainfall trend and variability	Is rain decreasing, and is it becoming less reliable?	Yearly rainfall totals over time, and how much they vary from year to year
Extreme rain	Risk of heavy storms and erosion	Months with unusually high rainfall

Signal	Meaning	Calculated from
Heat-stress days	Days that are too hot for crops	Daily maximum temperature above a crop-specific limit
Frost risk	Chance of damaging cold	Days with minimum temperature near or below freezing
Bare-field months	When the land has no plant cover (erosion and water loss risk)	Months where NDVI stays below a low threshold
Crop stress periods	When current crops are most vulnerable	NDVI drops, dry soil, or heat during the growing season
Water balance indicator	Is there enough water for the crop?	Rainfall compared with estimated evaporation and crop water need
Slope / erosion risk	Where soil is most likely to wash away	Elevation-based slope plus rainfall intensity

Simple definitions for the team

- **NDVI:** a number showing how green and healthy plants are. Higher means more green plants; near zero means bare soil.
- **Evapotranspiration:** water lost from soil and plants to the air. It tells us how much water crops really need.
- **Anomaly:** how far a value is from its normal long-term average.

6. Step 5: Rotation Strategies and Scoring

Crop facts table (built by us)

We create a small table with one row per crop and the properties the scoring needs. The values must be checked against FAO or local agricultural sources; the example below is only a starting idea.

Crop (example)	Water need	Nitrogen fixing	Soil cover / roots	Main use
Wheat	Medium	No	Seasonal cover	Food, income
Maize	High	No	Seasonal cover	Food, feed
Rice	Very high	No	Flooded field	Food (water-heavy)
Berseem clover	Medium	Yes	Good cover, adds organic matter	Animal forage
Faba beans	Low to medium	Yes	Moderate	Food, soil nitrogen
Sorghum	Low	No	Deep roots, drought tolerant	Feed, food

Building the options

Each option is a sequence of crops over a few seasons or years. The tool always includes:

- **Option A: Current rotation** (entered by the farmer), used as the baseline.
- **Option B:** a rotation that adds a nitrogen-fixing crop (for soil health).
- **Option C:** a rotation built around lower water use, such as a drought-tolerant crop and a cover crop.

Options can be built from simple rules (for example, never repeat the same crop twice in a row, avoid heavy water crops when water stress is high).

Scoring method (explainable on purpose)

1. Pick the criteria: **water use, soil health, erosion reduction, pest-cycle breaking, income potential, and fit with local climate**.
2. Give every option a score from 0 to 1 on each criterion, using the signals from Step 4 and the crop table. Example: if water stress is high, an option with a high-water crop scores low on water use.
3. Convert the farmer's priority sliders into **weights** that add up to 100%. A farmer who stresses saving water gives that criterion a bigger weight.
4. The final score of an option is the **weighted average** of its criterion scores.
5. Rank the options, but always show the individual criterion scores so the farmer sees *why*.

We use a transparent scoring system instead of a black-box model because the challenge asks us to explain why each strategy is suggested and what evidence supports it. Machine learning can play a supporting role (for example, grouping fields into zones with similar conditions, or detecting trends in time series).

7. Step 6: Uncertainty and Explanation

Uncertainty (required by the challenge)

Every option gets a **confidence level** (for example high, medium, low) and a list of **information gaps**. Confidence goes down when:

- Many satellite values were missing or filled in (clouds, gaps).
- A coarse dataset was stretched to a finer grid.
- Local soil data was missing and a default was used.
- Different datasets disagree with each other.
- The crop facts come from general sources and not from local trials.

Plain-language explanation

For each option the tool writes a short text covering: **why** it was suggested, **evidence** (which signals support it), **benefits**, **information gaps**, and **risks needing local review** (for example, seed availability or local market). If we use an LLM to write these texts, it must receive only our calculated numbers and must not add facts of its own. Every result also carries a reminder that the tool supports, and does not replace, local expertise.

8. Step 7: The App

Suggested architecture

Part	Suggested tools	Job
Data processing	Python (pandas, numpy, xarray, rasterio or similar)	Download, harmonize, and calculate signals offline; save clean results
Scoring engine	Python	Build rotations, score them, compute confidence
Backend	FastAPI	Receive the farmer's inputs, return options, scores, and explanations
Frontend	Web app (React or simple HTML), Leaflet or similar for maps, a charting library	Map, input forms, sliders, charts, comparison screen

How the farmer uses it (screens)

Screen	Farmer does	Notes
1. Location	Clicks the farm on the map	Required
2. Current rotation	Picks crops and their order	Required; becomes Option A
3. Soil and limits	Soil type, water availability, equipment, market	Optional; defaults used if skipped and shown as gaps
4. Priorities	Moves sliders: save water, income, soil health, erosion, pests	Become the scoring weights
5. Results	Sees 3+ options side by side with scores, explanations, confidence, and charts	Includes bare-field months, soil moisture and rainfall trends

Design rules: few required inputs, sliders and dropdowns instead of typing, simple words, and Arabic plus English if time allows.

9. Suggested Team Roles

Adjust to the real team size; one person can take several roles.

Role	Responsibilities
Data lead	Pilot area, downloading datasets, harmonization, data log
Analysis / scoring lead	Signals, crop facts table, rotation options, scoring, confidence
Backend developer	API that connects the data and scoring to the app
Frontend / design lead	Map, input screens, results screen, farmer-friendly wording
Presentation and research lead	Project description, demo story, references, limitations, NASA data explanation

10. MVP Priorities and Limitations

Finish in this order

1. One pilot region with precomputed, harmonized data.
2. Signals: soil moisture anomaly, rainfall trend, heat and frost days, bare-field months.
3. 3 rotation options with scoring and explanations.
4. Map plus comparison screen with confidence and information gaps.
5. Bonus: rainfall/temperature scenario sliders, GRACE and ECOSTRESS, Arabic interface.

Limitations to state honestly in the presentation

- Satellite data is coarse compared with a single field and is affected by clouds.
- Crop facts are general, not local trial results.
- The tool shows patterns and possibilities, not guaranteed outcomes.
- Local farmers and agricultural experts must review every recommendation.